

EXPERIMENT-27 Best First Search

Aim

To implement the **Best First Search algorithm** using Prolog.

Objectives

- To understand heuristic search.
- To select the node with the lowest heuristic value.
- To find a path from a start node to a goal node.

Algorithm

1. Define the graph.
2. Define heuristic values for nodes.
3. Start with the initial node.
4. Select the node having the smallest heuristic value.
5. Expand its neighboring nodes.
6. Continue until the goal node is reached.
7. Display the path.

Flowchart

START

|

v

Put start node

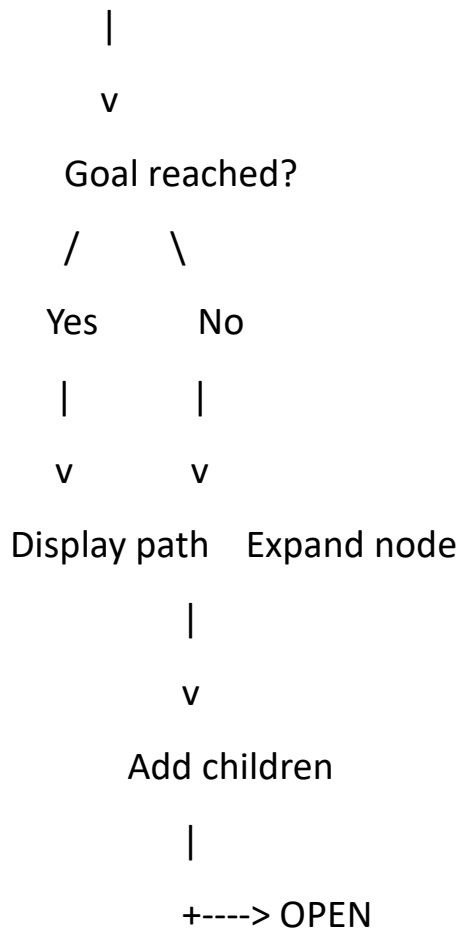
in OPEN list

|

v

Select node with minimum

heuristic value



Program

% Best First Search

edge(a, b).

edge(a, c).

edge(b, d).

edge(b, e).

edge(c, f).

edge(c, g).

edge(e, h).

edge(f, h).

heuristic(a, 6).

heuristic(b, 4).

heuristic(c, 5).

heuristic(d, 7).

heuristic(e, 2).

heuristic(f, 3).

heuristic(g, 8).

heuristic(h, 0).

best_first(Start, Goal, Path) :-

 search([Start], Goal, [Start], RevPath),

 reverse(RevPath, Path).

search([Goal|_], Goal, Path, Path).

search([Current|Rest], Goal, Visited, Path) :-

 findall(H-Next,

 (edge(Current, Next),

 \+ member(Next, Visited),

 heuristic(Next, H)),

 Children),

 sort(Children, Sorted),

 extract_nodes(Sorted, Nodes),

 append(Nodes, Rest, NewOpen),

 NewVisited = [Current|Visited],

 search(NewOpen, Goal, NewVisited, Path).

```
extract_nodes([], []).
```

```
extract_nodes([_Node|T], [Node|R]) :-
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```
    extract_nodes(T, R).
```

Query

```
?- best_first(a, h, Path).
```

Output

File	I/O	AI Agent
HelloWorld.pl	STDIN Input for the program (Optional)	
<pre>1 % Best First Search 2 3 % Graph edges 4 edge(a, b). 5 edge(a, c). 6 edge(b, d). 7 edge(b, e). 8 edge(c, f). 9 edge(c, g). 10 edge(e, h). 11 edge(f, h). 12 13 % Heuristic values 14 heuristic(a, 6). 15 heuristic(b, 4). 16 heuristic(c, 5). 17 heuristic(d, 7). 18 heuristic(e, 2). 19 heuristic(f, 3). 20 heuristic(g, 8). 21 heuristic(h, 0). 22 23 % Best First Search 24 best_first(Start, Goal, Path) :- 25 search([Start], Goal, [Start], RevPath), 26 reverse(RevPath, Path). 27 28 % Goal reached 29 search([Goal _], Goal, Path, Path). 30</pre>		Output: Best First Search Path: [a,a,b,e]

Result

Best First Search was successfully implemented using heuristic values.

Conclusion

Best First Search selects the most promising node according to its heuristic value.